

Key Concepts, *Homeostasis* & Hormesis.

Course	GSM · Friedrich-Schiller-Universität Jena
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Four ideas that frame everything that follows.

01

Activity & Energy Cost

What we mean by physical activity, exercise, training and sport — and how each shows up in the energy budget.

02

Acute Homeostasis

Negative feedback that rebalances the internal environment within seconds when exercise perturbs it.

03

Hormesis & Adaptation

Why the same stressors that disturb homeostasis become — at the right dose — the trigger for long-term gains.

DEFINITIONS MATTER

Physical activity, exercise, training, sport — *related but not interchangeable.*

BROADEST Physical activity Any bodily movement produced by skeletal muscles that results in energy expenditure.	PLANNED Exercise Planned, structured, repetitive activity with the explicit goal of maintaining or improving fitness.	SYSTEMATIC Training Systematic exercise designed to produce long-term physiological adaptation.	COMPETITIVE Sport Rule-bound, competitive activity with explicit performance objectives.
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The more intensive the physical activity, the more energy is consumed per unit of time.

TOTAL DAILY ENERGY EXPENDITURE

The day's energy bill is the sum of *four components* — only one of which we choose.

$$TDEE = BMR + TEF + AEE + NEAT$$

BMR	TEF	AEE	NEAT
Basal metabolic rate	Thermic effect of food	Activity energy expenditure	Non-exercise thermogenesis
Energy required for essential physiological function at complete rest — the largest piece of the budget.	Energy spent digesting, absorbing and processing nutrients — small, but never zero.	Energy consumed during deliberate physical activity — the variable we can program.	Fidgeting, posture, walking to the kettle — surprisingly large in active lifestyles.

STRAIN DEPENDS ON	Type	Intensity	Duration & frequency	Individual factors
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WHEN EXERCISE BEGINS

Five physiological systems shift in concert to *protect homeostasis*.

CARDIOVASCULAR

Pump & redistribute

Up, to deliver O₂ & substrates

Up — more blood per beat

20–25 % → 85–90 % to muscle

Systolic rises > diastolic

RESPIRATORY

Ventilate harder

Up — more breaths per minute

TIDAL VOLUME

Up — deeper breaths

Scales with intensity

More O₂ extracted from each breath

METABOLIC

Re-fuel the contraction

ATP DEMAND

Surges with workload

SUBSTRATE

Fat → CHO as intensity rises

LACTATE

Rises beyond anaerobic threshold

HORMONES

Catecholamines, cortisol, glucagon

Shed the heat

Rises with metabolic load

Primary heat-loss mechanism

Up — convective dissipation

Held within narrow band

IMMUNEREGULATION

Mobilise & survey

Transient rise in circulation

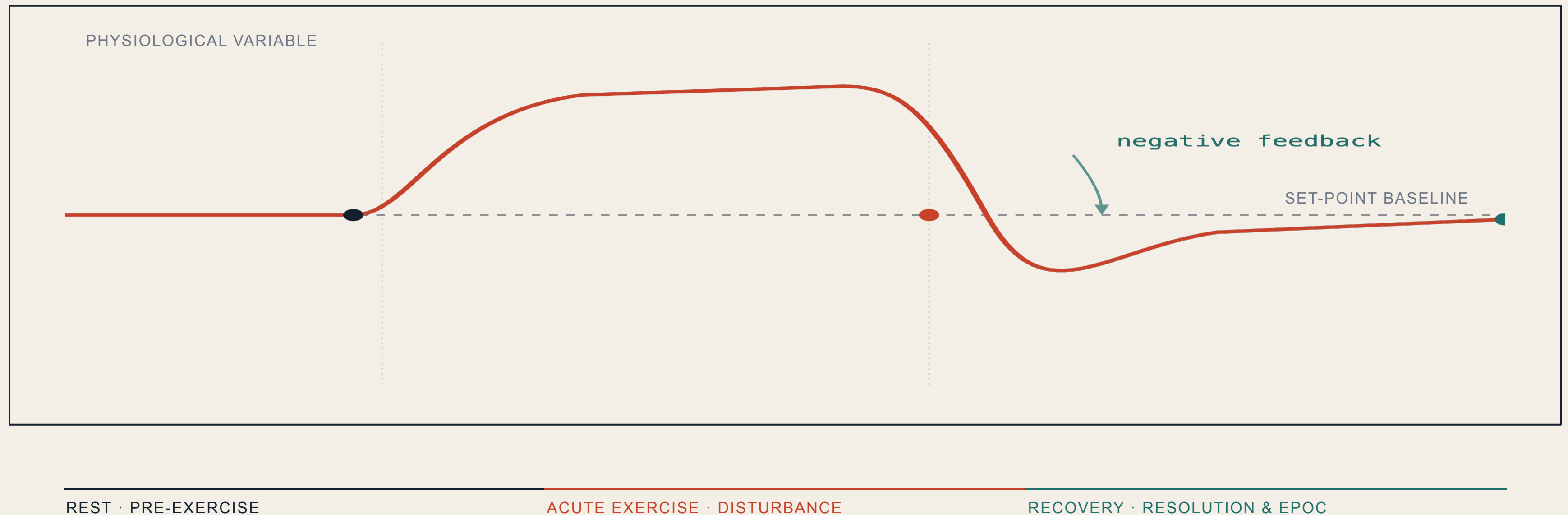
Released from working muscle

Cells migrate to tissues

IL-10, IL-1Ra follow the peak

THE TIME-COURSE OF RECOVERY

Exercise displaces the set-point; *negative feedback* brings it home.



A BIPHASIC DOSE-RESPONSE

Below threshold there is no signal; above it there is harm — adaptation lives in *the middle zone* .

ZONE 1 · SUB-THRESHOLD Sedentary → light Insufficient stimulus to engage adaptive programmes. URTI risk near the population mean. No meaningful adaptation	ZONE 2 · HORMETIC Moderate → vigorous Net adaptive benefit peaks. Nrf2, PGC-1α, HSP and autophagy pathways maximally engaged. URTI risk ↓ ~38 % vs baseline	ZONE 3 · OVERLOAD Extreme & sustained Inflammatory load overwhelms adaptive reserve; immune suppression and the open-window effect. URTI risk > sedentary baseline
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Adaptive triggers — IL-6, ROS, NF-κB — are the same molecules that signal stress. Magnitude and duration decide the outcome.

J-CURVE · NIEMAN 1994
CALABRESE & MATTSON 2017

SAME BIOLOGY, TWO TIMESCALES

Homeostasis governs *resolution* ; hormesis governs *adaptation*.

DIMENSION	HOMEOSTASIS · 0–24 H	HORMESIS · 24 H → CYCLES
ROLE OF IL-6 / ROS	Error signal — magnitude of perturbation	Adaptive trigger — dose-dependent
FEEDBACK DIRECTION	Negative · corrective	Forward at low dose · negative at overload
SET-POINT OUTCOME	Preserved at baseline	Elevated · supercompensation
DOSE-RESPONSE SHAPE	Linear, monotonic resolution	Biphasic, inverted-U
KEY REGULATORS	IL-10 axis · HPA cortisol	Nrf2 · PGC-1α · HSPs · autophagy
FAILURE MODE	Unresolved chronic inflammation	Overtraining · immunosuppression

| *Blunting inflammation post-exercise (NSAIDs, antioxidants) can suppress the very signal training depends on.*

SLEEP ARCHITECTURE & EXERCISE

Sleep is when the body *cashes in* the day's adaptive signal.

NREM · 1	Light sleep Transition from wakefulness; muscle tone falls.	~5 %
NREM · 2	Intermediate Sleep spindles, K-complexes; further wave slowing.	~45 %
NREM · 3	Deep / SWS Highest arousal threshold; physical restoration.	~25 %
REM	Active sleep Brain near-wakeful; muscles atonic; dream-rich.	20–25 %

EXERCISE → SLEEP

Activity earns deeper rest

↑ adenosine → sleep pressure

Core-temp drop → faster onset

↓ cortisol & psychological stress

Circadian re-alignment

SLEEP → EXERCISE

Rest unlocks the gain

Strength & power preserved

Lower injury risk

Mitochondrial & vascular adaptation

Immune competence & insulin sensitivity

WHERE THE FOUR THREADS MEET

Activity perturbs, recovery resolves, dose decides —
and *sleep books the gain*.

01 · Activity

Words have scopes — use them carefully.

Activity, exercise, training, sport sit on a ladder of intent and structure. Energy expenditure is the one currency they all share.

02 · Two frameworks

Homeostasis resolves; hormesis adapts.

Negative feedback restores the baseline within 24 h. The same inflammatory signal — at the right dose — lifts the set-point over weeks.

03 · Recovery

Sleep is part of the prescription.

7–9 h, with intact NREM-3 and REM, is when the molecular memory of training is consolidated. Skip it and the dose-response collapses.

Next: *Immunity, limits of performance & energy balance* — how the systems we just framed cash out in muscular work.